

FORKLIFT BATTERY CHARGING

Manage hydrogen, acid splash, electrical arcing, heavy battery and connection hazards from lead-acid traction batteries through

PURPOSE AND SCOPE

Manage hydrogen, acid splash, electrical arcing, heavy battery and connection hazards from lead-acid traction batteries through designated charging areas, ventilation, correct equipment and emergency flushing.

KEY HAZARDS

1  HYDROGEN	2  ACID SPLASH	3  ELECTRICAL ARC	4  HEAVY BATTERY
5  WRONG POLARITY	6  DAMAGED LEAD	7  IGNITION SOURCE	8  EYE CONTACT

CRITICAL VALUES TO VERIFY IN THE FIELD

0 Spark, flame or smoking in charging area	100% Charger-battery voltage/connector match	Immediate access Eye/face wash and spill kit
--	--	--



REMEMBER - SERNEM

- S** Stop and think
- E** Evaluate the risk
- R** Remove or reduce
- N** Never assume
- E** Ensure protection
- M** Monitor and communicate

FORKLIFT BATTERY CHARGING

Manage hydrogen, acid splash, electrical arcing, heavy battery and connection hazards from lead-acid traction batteries through

STANDARDS AND FIELD REQUIREMENTS

OSHA 1910.178(g), 1910.151(c) and manufacturer instructions address designated charging areas, ventilation, fire prevention, battery handling and emergency flushing.

OSHA 1910.178(g)(1)-(2)

CHARGING AREA AND EQUIPMENT

Charge in designated areas with arrangements for battery handling, positioning, protection and ventilation.

OSHA 1910.178(g)(5)-(7)

ACID AND IGNITION CONTROL

Provide neutralization/cleanup for electrolyte and prohibit smoking, open flames and sparks.

OSHA 1910.178(g)(8)-(12)

SAFE CONNECTION

Charge with truck braked and off, open covers, keep metal objects away and do not disconnect under load.

OSHA 1910.151(c)

EMERGENCY FLUSHING

Provide quick drenching or flushing facilities where corrosive electrolyte exposure can occur.

MINIMUM EVIDENCE BEFORE START

- | | |
|---|--|
| ☐ Charging-area risk assessment | ☐ Ventilation inspection |
| ☐ Charger/battery compatibility record | ☐ Lead/connector inspection |
| ☐ Eyewash test record | ☐ Electrolyte spill response plan |

SUPERVISOR DISCUSSION POINTS

- Where can hydrogen accumulate?
- What is the disconnection sequence?
- What is the first action for acid in the eye?

VENTILATED CHARGING AREA AND ACID EMERGENCY READINESS



Correct charger - Ventilation - No sparks - Face shield - Eyewash - Kit

FORKLIFT BATTERY CHARGING

Manage hydrogen, acid splash, electrical arcing, heavy battery and connection hazards from lead-acid traction batteries through

CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

1. SELECT SEALED / LOW-MAINTENANCE SYSTEM

Use sealed batteries or automatic watering where suitable.

2. VENTILATED CHARGING AREA

Charge in a ventilated, impact-protected area with ignition sources removed.

3. CORRECT CHARGER AND CONNECTION

Verify voltage, polarity and connector; switch off the charger before connection or disconnection.

4. ACID PPE AND EMERGENCY FLUSHING

Wear face shield, acid gloves and apron; keep eyewash and neutralization kit accessible.

CRITICAL WARNING



A spark can ignite gas released during charging. Never disconnect a live connector or bring metal tools/jewelry near terminals.

SITE CHECKLIST (BEFORE STARTING WORK)

- | | |
|--|--|
| ☐ Area ventilated | ☐ No ignition source |
| ☐ Truck off/braked | ☐ Charger matches battery |
| ☐ Cable/connector sound | ☐ Covers open |
| ☐ Acid PPE worn | ☐ Eyewash ready |
| ☐ Spill kit ready | |



FORKLIFT BATTERY CHARGING

Manage hydrogen, acid splash, electrical arcing, heavy battery and connection hazards from lead-acid traction batteries through

ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			